

MANGALORE REFINERY AND PETROCHEMICALS LIMITED (MRPL)

REFINERY BLOCK 3 · CRUDE DISTILLATION UNIT II (CDU-II)

NON-DESTRUCTIVE TESTING (NDT) & ROTATING EQUIPMENT INSPECTION REPORT

1. GENERAL ASSET & INSPECTION METADATA

Equipment Identifier: P-204 A/B (Heavy Gas Oil Transfer Pump)
Service Fluid: Heavy Gas Oil (HGO) @ 360°C Operating Temp, 24 kg/cm²g
Plant Unit: CDU-II Primary Fractionation Section
Inspection Date: 2026-08-14
Inspector Lead: Rajesh K. Sharma (Level-II NDT Inspector)
Inspection Method: Ultrasonic Thickness Gauging (UTG) & Vibration Spectrum Analysis

2. NON-DESTRUCTIVE TESTING (NDT) MEASUREMENTS & FINDINGS

Parameter	Measured Value	Design / Limit	Assessment
Nominal Wall Thickness	8.2 mm	8.2 mm (Sched 80 CS)	Baseline
Minimum Wall Thickness (Discharge Bend)	4.2 mm	4.0 mm Retirement (API 570)	CRITICAL (Near Retirement)
Corrosion Rate (High Temp Sulfidic)	0.8 mm/year	0.15 mm/year Design Limit	HIGH / ACCELERATED
Overall Vibration (Pump DE Horizontal)	7.5 mm/s RMS	4.5 Alarm / 7.1 Trip (ISO)	CRITICAL (Zone D Exceeded)
Pump Bearing Temperature	88.4 °C	85.0 °C Maximum Continuous	WARNING (Elevated)

3. INSPECTION OBSERVATIONS & DEFECT LOG

- Severe localized wall thinning detected at the 90-degree discharge elbow bend downstream of P-204.
- Thickness measured at 4.2 mm, indicating only 0.2 mm remaining margin above API 570 retirement (4.0 mm).
- Vibration spectrum indicates significant 1x RPM shaft unbalance and 2x blade pass frequency spikes.
- Inboard bearing audible distress noted during run; cavitation surging observed at suction strainer.
- Risk of catastrophic pressurized hydrocarbon leak if run continues without immediate containment.

4. OFFICIAL INSPECTOR ENDORSEMENT

Recorded By: Rajesh K. Sharma, Level-II NDT Inspector (NDT-CERT-MRPL-4092)

Disposition: **CONDITIONAL** · Immediate Spool Wrap or Replacement Mandated before return to run.

Document Reference: MRPL/CDU2/INSP/2026/08/P204